

*Example 2.8 (Beeman's Algorithm)* The method treats the positions and momenta differently, updating these from the formulas

$$\mathbf{q}_{n+1} = \mathbf{q}_n + h\dot{\mathbf{q}}_n + \frac{h^2}{6}[4\ddot{\mathbf{q}}_n - \ddot{\mathbf{q}}_{n-1}]. \quad (2.31)$$

$$\mathbf{p}_{n+1} = \mathbf{p}_n + \frac{h}{6}\mathbf{M}[2\ddot{\mathbf{q}}_{n+1} + 5\ddot{\mathbf{q}}_n - \ddot{\mathbf{q}}_{n-1}]. \quad (2.32)$$

The shorthand  $\dot{\mathbf{q}}_n \equiv \mathbf{M}^{-1}\mathbf{p}_n$ ,  $\ddot{\mathbf{q}}_n \equiv \mathbf{M}^{-1}\mathbf{F}(\mathbf{q}_n)$  has been used.

This method requires that the positions (and forces) be known at two successive points  $h$  apart in time in order to initialize the iteration. These might be generated by using the Verlet method or some other self-starting scheme. Beeman's algorithm is explicit since, given  $\mathbf{q}_n$ ,  $\mathbf{q}_{n-1}$  and  $\mathbf{p}_n$ , one directly obtains  $\mathbf{q}_{n+1}$  and then,  $\ddot{\mathbf{q}}_{n+1}$ , and thus  $\mathbf{p}_{n+1}$ , with only one new force evaluation. Because it is a "partitioned multistep method," its analysis is more involved than for the one-step methods, and, in particular its qualitative features are difficult to relate to those of the flow map. The order of accuracy of the scheme above can be shown to be three.

## Exercises

1. The Morse potential is often encountered in molecular modelling and takes the form

$$\varphi_{\text{Morse}}(r) = D(1 - e^{-a(r-r_e)})^2.$$

Consider a one-particle oscillator in the Morse potential (Hamiltonian  $H = p^2/2 + \varphi_{\text{Morse}}(q)$ ). Explain why the trajectory of a particle will be bound for all time if the initial energy  $H(q_0, p_0) < D$ . Implement the 4th order Runge-Kutta and Verlet methods and explore the problem for  $a = 1, D = 1, r_e = 1$ . You should observe that all trajectories of the Runge-Kutta method eventually diverge. Study the dissociation time (the time until the particle is ejected to some fixed large distance, say  $r = 3$ ) as a function of the stepsize and the initial energy.

2. Demonstrate the bound (2.12). [Hint: use  $e^{Lh} > 1 + hL$ .]
3. Let  $f$  satisfy the Lipschitz condition  $|f(u) - f(v)| \leq L|u - v|$ . Let  $\mathcal{G}_h$  represent the flow map approximation associated to Euler's method. Show that

$$|\mathcal{G}_h(u) - \mathcal{G}_h(v)| \leq (1 + hL)|f(u) - f(v)|.$$

Next prove (2.10) holds by finding  $\Delta$ ,  $\bar{K}$  and  $p$ . This can be used to prove Theorem 2.1.